1. Incompatible Data Types — Common Errors

Operation	Error	Why?
10 + "5"	TypeError	int and str cannot be added directly.
"10" - 5	TypeError	str and int subtraction is unsupported.
"Hi" * 2.5	TypeError	String repetition requires an integer count.
10 + None	TypeError	NoneType and int addition is unsupported.
10 + [20]	TypeError	int and list cannot be added.
[1, 2] + (3, 4)	TypeError	list and tuple cannot be concatenated directly.
{1, 2} + {3, 4}	TypeError	Sets do not support +.
{"a": 1} + {"b": 2}	TypeError	Dictionaries do not support +.
10 < "20"	TypeError	Integers and strings cannot be ordered against each other.
len(10)	TypeError	An integer has no length.
list(10)	TypeError	An integer is not iterable.
int(None)	TypeError	None cannot be directly converted to an integer.
int("hello")	ValueError	The string is not a valid integer representation.
int("3.5")	ValueError	int() does not parse a decimal-formatted string.

2. Common combinations that work

Har operation between different types error nahi deta.

Operation	Output	Reason
10 + 2.5	12.5	Compatible numeric types
"Hi" + " Bro"	"Hi Bro"	String concatenation
"Hi" * 3	"HiHiHi"	String repetition
[1, 2] + [3]	[1, 2, 3]	List concatenation
(1, 2) + (3,)	(1, 2, 3)	Tuple concatenation
10 == "10"	False	Equality comparison is allowed
True + 2	3	bool behaves as a subtype of int
10 / 2	5.0	True division returns a float

3. Important distinction: TypeError vs ValueError

- **TypeError:** Operation ke liye data type compatible nahi hai. Example: 10 + "5".
- **ValueError:** Input type acceptable hai, lekin uski value required format mein nahi hai. Example: int("hello").

4. One important rule

Python har incompatible operation ko automatically convert nahi karta.

age = "21"print(age + 5) # **TypeError**

Correct approach depends on your intention:

print(int(age) + 5) # 26: numeric additionprint(age + str(5)) # 215: string concatenation

Interview tip: Incompatible data types ko study karte waqt operators (+, -, *, /, comparisons), built-in functions (len(), int(), list()), aur type conversion errors ko alag-alag practice karo.